Problem I. Incyde

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Vanessa has n memories about humanity, numbered $0, 1, \cdots, n-1$, arranged in this order on a circle. *Vanessa* divides them into m contiguous intervals, computes the sum S_i of the elements in each interval, and defines the weight of the division as the greatest common divisor of all the S_i.

Through the Eye of *Horus*, *Vanessa* sees all $\binom{n}{m}$ possible divisions, but she cannot endure so many memories. You need to help her calculate the **product** of the weights of all divisions. Since the answer may be large, output it modulo $10^9 + 123$.

Recall that a positive integer is a common divisor of a set S if and only if it divides every element of S.

Input

This problem contains multiple test cases.

The first line contains a positive integer T ($1 \le T \le 100$), representing the number of test cases.

For each test case, the only line contains two integers n, m ($2 \le m \le n \le 10^{12}$).

Output

For each test case, output one line containing an integer representing the answer.

Example

standard input	standard output
8 4 2 5 2 10 3 2026 1018 1000000123 114514 2000000247 1000000123 67325072148 1145141919 999000122877 1997	108 400 895412257 711051382 1 52312939 472236908 0

Note

For the first test case, the weights of all six divisions are listed below:

- $\gcd(0, 1+2+3) = 6$
- $\gcd(0+1, 2+3) = 1$
- $\gcd(0+1+2, 3) = 3$
- $\gcd(1, 2+3+0) = 1$
- $\gcd(1+2, 3+0) = 3$
- $\gcd(2, 3+0+1) = 2$

Therefore, the answer is $6 \cdot 1 \cdot 3 \cdot 1 \cdot 3 \cdot 2 = 108$.